Experiment 9: AC Three-Phase Circuits with Balanced Load

OBJECTIVE:

1. To understand balanced three-phase voltages.
2. To measure the phase and line voltages of a balanced connected load.
3. To measure the phase and line current of a balanced connected load.

INTRODUCTION:

Three Phase System

A 3-Φ system is formed by 3*1-Φ systems, which are physically connected together for 3-Φ operation. To maintain constant power flow, the three single phase circuits must be operated with the same voltage and current magnitudes in each phase (circuit), with a relative phase angle of 120 electrical degrees apart.

A typical three-phase system consists of three voltage sources connected to loads by three or four wires (or transmission lines). A three-phase system is equivalent to three single-phase circuits. The voltage sources can be either wye-connected as shown in Fig. 1(a) or delta-connected as in Fig. 1(b).

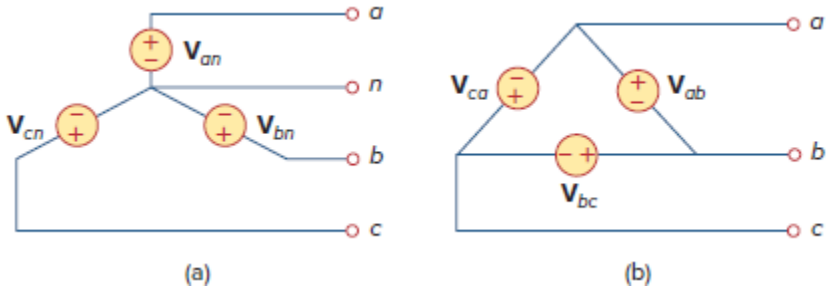


Figure 1: Three-phase voltage sources: (a) Y-connected source, (b) Delta Δ-connected source.

Balanced phase voltages are equal in magnitude and are out of phase with each other by 120°.

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$
$$\mathbf{V}_{bn} = V_p \angle -120^\circ$$
$$\mathbf{V}_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

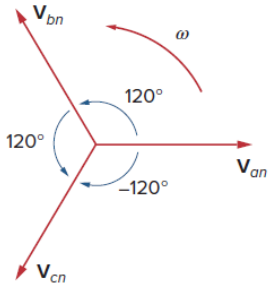
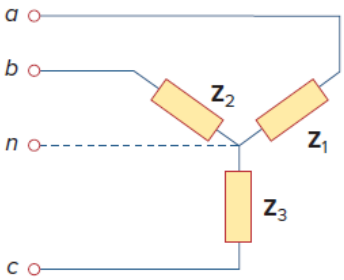


Figure 2: Phase sequences: abc or positive sequence

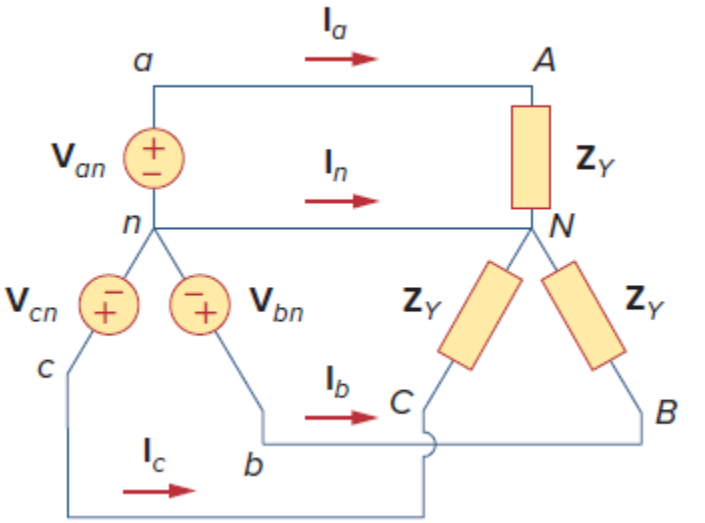
where Vp is the effective or rms value of the phase voltages. This is known as the abc sequence or positive sequence. In this phase sequence, Van leads Vbn, which in turn leads Vcn.

A balanced load is one in which the phase impedances are equal in magnitude and in phase. For a balanced wye-connected load,



$$\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}_3 = \mathbf{Z}_Y$$

Figure 3: a Y-connected load



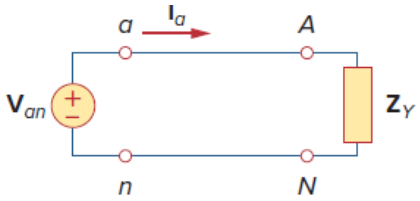
$$\mathbf{V}_{an} = V_p \angle 0°$$

$$\mathbf{V}_{bn} = V_p \angle -120°, \qquad \mathbf{V}_{cn} = V_p \angle +120°$$

Figure 4: Balanced Y-Y connection

the magnitude of the line voltages V_L is √3 times the magnitude of the phase voltages Vp, or

$$V_L = \sqrt{3} V_p$$



$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

Figure 5: A single phase equivalent circuit

Summary of phase and line voltages/currents for balanced three-phase systems.[1]

Connection	Phase voltages/currents	Line voltages/currents
Y-Y	$\mathbf{V}_{an} = V_p\angle 0°$ $\mathbf{V}_{bn} = V_p\angle -120°$ $\mathbf{V}_{cn} = V_p\angle +120°$ Same as line currents	$\mathbf{V}_{ab} = \sqrt{3}V_p\angle 30°$ $\mathbf{V}_{bc} = \mathbf{V}_{ab}\angle -120°$ $\mathbf{V}_{ca} = \mathbf{V}_{ab}\angle +120°$ $\mathbf{I}_a = \mathbf{V}_{an}/\mathbf{Z}_Y$ $\mathbf{I}_b = \mathbf{I}_a\angle -120°$ $\mathbf{I}_c = \mathbf{I}_a\angle +120°$
Y-Δ	$\mathbf{V}_{an} = V_p\angle 0°$ $\mathbf{V}_{bn} = V_p\angle -120°$ $\mathbf{V}_{cn} = V_p\angle +120°$ $\mathbf{I}_{AB} = \mathbf{V}_{AB}/\mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{BC}/\mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{CA}/\mathbf{Z}_\Delta$	$\mathbf{V}_{ab} = \mathbf{V}_{AB} = \sqrt{3}V_p\angle 30°$ $\mathbf{V}_{bc} = \mathbf{V}_{BC} = \mathbf{V}_{ab}\angle -120°$ $\mathbf{V}_{ca} = \mathbf{V}_{CA} = \mathbf{V}_{ab}\angle +120°$ $\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30°$ $\mathbf{I}_b = \mathbf{I}_a\angle -120°$ $\mathbf{I}_c = \mathbf{I}_a\angle +120°$
Δ-Δ	$\mathbf{V}_{ab} = V_p\angle 0°$ $\mathbf{V}_{bc} = V_p\angle -120°$ $\mathbf{V}_{ca} = V_p\angle +120°$ $\mathbf{I}_{AB} = \mathbf{V}_{ab}/\mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{bc}/\mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{ca}/\mathbf{Z}_\Delta$	Same as phase voltages $\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30°$ $\mathbf{I}_b = \mathbf{I}_a\angle -120°$ $\mathbf{I}_c = \mathbf{I}_a\angle +120°$
Δ-Y	$\mathbf{V}_{ab} = V_p\angle 0°$ $\mathbf{V}_{bc} = V_p\angle -120°$ $\mathbf{V}_{ca} = V_p\angle +120°$ Same as line currents	Same as phase voltages $\mathbf{I}_a = \dfrac{V_p\angle -30°}{\sqrt{3}\mathbf{Z}_Y}$ $\mathbf{I}_b = \mathbf{I}_a\angle -120°$ $\mathbf{I}_c = \mathbf{I}_a\angle +120°$

EQUIPMENT AND COMPONENT

1. AC Power source
2. Impedance load
3. Multimeter
4. 4 Channel Oscilloscope

PROCEDURE



Figure 6: Multisim 14.0 software icon

1. Run Multisim 14.0 Software.
2. Click File > Save As > EEE212_Experiment 6_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 9 to construct your balanced Y-Y connection circuit as shown in figure 7.

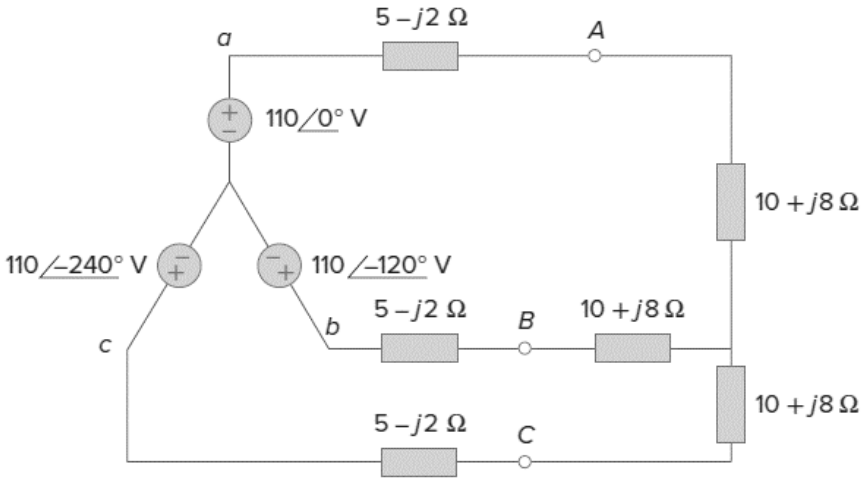


Figure 7: Three-wire Y-Y system

4. Click Place > Component > Group > Sources > POWER_SOURCES > AC_POWER > OK. Place it on the screen.
5. Set your AC_POWER to 110 V_{rms}, Freq. = 50Hz, Phase = 0° for *line a*.

6. Repeat step 5 to change another phase voltage to -120° and +120° with the same magnitude of ac voltage.
7. Set your line impedance, Z_l = 5-j2 Ω for all line impedance.
8. Set your Load impedance, Z_L = 10+j8 Ω for all load impedance.
9. Connect all component as shown in Figure 7.
10. Run the simulation

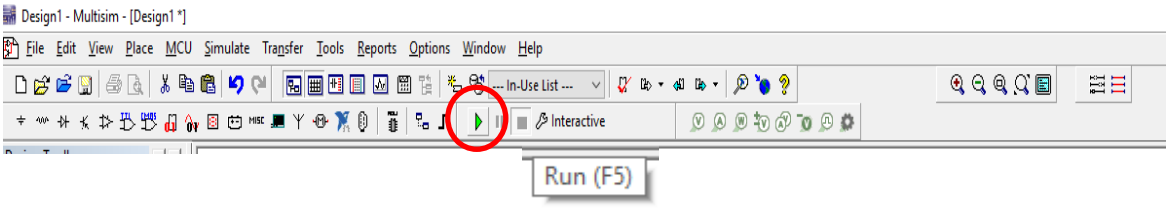


Figure 8: Run button/Icon

11. Measure the line voltage, phase voltage, line current and phase current. Record your answer in Table 1.
12. Repeat the same procedure, repeat for Y-Δ, by using the same balanced load.

RESULT

Table 1 (Y-Y)

	Measured value	Calculated value
Line Current (A)		
Phase current (A)		
Phase voltage, Vp		
Line Voltage, V_{line}		

Table 2 (Y-Δ)

	Measured value	Calculated value
Line Current (A)		
Phase current (A)		
Phase voltage, Vp		
Line Voltage, V_{line}		

DISCUSSION

The following points should be discuss in your report.

1. Discuss briefly your finding in the experiment.
2. Discuss briefly the advantages of 3-phase system compare to single phase.
3. Show your working calculation for Table 1 and Table 2.